

SUJITH SRINIVAS GRANDHE

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Software Engineer · Java · Spring Boot · AWS · Kafka · Kubernetes · Distributed Systems · AI/LLM Workflows

PROFESSIONAL SUMMARY

Software Engineer with 6 years of experience designing and building high-throughput, low-latency backend platforms using Java, Spring Boot, microservices, and cloud-native development on AWS and Kubernetes. Proven expertise optimizing real-time systems processing 22K+ transactions/sec at sub-100ms latency, sustaining 99.99% availability, and cutting cloud spend by \$22K/month across fraud, risk, underwriting, inventory, and e-commerce decisioning workloads. Skilled in event-driven data processing (Kafka Streams, Kinesis), API design (REST, gRPC, GraphQL federation), CI/CD automation with integrated security scanning (Jenkins, GitHub Actions, Snyk), and full-stack development (Angular, React, D3.js). Strong background in scaling distributed systems, migrating legacy monoliths to AWS EKS, and building data pipelines and ETL workflows for analytics and reporting. Hands-on experience integrating AI/ML, RAG-based LLM, and document intelligence workflows into fraud detection. Passionate about performance tuning, chaos engineering, observability (New Relic, AWS X-Ray), and mentoring engineering teams.

TECHNICAL SKILLS

Programming Languages: Java, Python, Node.js, C, C++, C#, Go, Kotlin, Groovy, Swift, JavaScript, TypeScript, SQL

Backend Engineering: Java 21/17/11, Spring Boot 3, Spring WebFlux, Spring Security, Spring Batch, Microservices, REST APIs, gRPC/Protocol Buffers, WebSockets, API Gateway, Service Discovery, Resilience4j, Serverless, Batch Processing, GraphQL Federation, Payment Gateway Integration, Distributed Systems

Cloud & DevOps: AWS — EC2, S3, RDS, Lambda, ECS, EKS, DynamoDB, CloudWatch, X-Ray, CodeDeploy, Textract, CloudFormation, AWS CDK; Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Infrastructure as Code, Canary Deployments, Rollback Strategies

Event Streaming & Messaging: Apache Kafka, Kafka Streams, AWS Kinesis, Event-Driven Architecture, Stream Processing, Retry/DLQ Patterns, Idempotent Consumers, Backpressure Handling, Fault-Tolerant Messaging

Databases & Caching: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Single-Table Design, Query Optimization, Indexing, Partitioning, Database Sharding, Caching Strategies

System Design & Architecture: High-Throughput Systems (22K+ TPS), Low-Latency Architecture, Scalability, Fault Tolerance, Idempotency, Circuit Breakers, Rate Limiting, Load Balancing, Monolithic, Microservices, MVC, Event-Driven Architecture, Monolithic-to-Microservices Migration

Reliability & Performance: SLA/SLO Optimization (99.99%), P99 Latency Optimization, Chaos Engineering (Gremlin), Incident Response & Resolution, Root Cause Analysis, Performance Tuning, Load Testing, JVM/GC Tuning, Cost Optimization

Observability & Monitoring: Prometheus, Grafana, ELK Stack, Splunk, New Relic APM, AWS X-Ray, Distributed Tracing, Centralized Logging, Log Correlation, MTTR Reduction

Security & API Engineering: OAuth2, JWT Authentication, Spring Security, API Security, Secure Data Handling, Compliance, OpenAPI/Swagger, SonarQube, Snyk Security Scanning

AI/ML & Data Engineering: Python, scikit-learn, Rule-Based Algorithms, Fraud Detection Models, Anomaly Detection, False-Positive Reduction, Regulatory Compliance Models, LLMs, RAG, LangChain, LangGraph, Document Intelligence (Textract), Data Pipelines, ETL, Transaction Intelligence

Frontend & Visualization: Angular, React.js, HTML5, CSS3, JavaScript, TypeScript, RxJS, State Management, D3.js, GraphQL, WebSockets, Real-Time Dashboards

Mobile/iOS Development: Swift, Xcode, SwiftUI/UIKit, URLSession, Codable, XCTest, TestFlight

Testing & Quality: Unit Testing, JUnit, Mockito, Postman, Selenium, JMeter, Locust, Test-Driven Development (TDD), Behavior-Driven Development (BDD), A/B Testing, Code Reviews, Refactoring, Debugging, Documentation (Swagger)

Methodology & Collaboration: SDLC, Agile, Scrum, Waterfall, OOP, SOLID Principles, Design Patterns, Pair Programming, Mentoring

Version Control & Tools: Git, GitHub, Bitbucket, SVN

PROFESSIONAL EXPERIENCE

Software Engineer | PNC Bank, USA

Feb 2026 – Present

- Engineered real-time fraud and risk decisioning microservices using Java 21, Spring Boot 3, AWS Lambda, ECS/EKS, and DynamoDB, processing **22K+** transactions/sec at **sub-100ms** latency while reducing false positives by **28%**.
- Implemented Kafka Streams event-processing pipelines across 15+ services for transaction ingestion, enrichment, and scoring, absorbing **3x** baseline volume during peak holiday card-authorization spikes with zero downtime via idempotent consumers, retry/backoff logic, and DLQ patterns.
- Optimized DynamoDB single-table design through advanced partitioning and access-pattern modeling, alongside PostgreSQL indexing and Redis caching layers, maintaining **<10ms P99** latency at **12K+** reads/sec.
- Migrated **3 legacy monoliths** to AWS EKS, achieving **99.99%** availability through pod autoscaling, health checks, circuit breakers, rate limiting, and rollback-ready canary releases.
- Designed a GraphQL federation layer aggregating **8+** banking APIs behind Spring Security, OAuth2, and JWT, reducing mobile client data usage by **25%** through query optimization and schema-level field pruning.
- Developed Angular dashboards with WebSocket integration to stream real-time fraud alerts to investigators, reducing investigation time by **35%**.
- Built AI-driven fraud and document intelligence workflows using Python, scikit-learn, RAG-based LLM patterns, and AWS Textract, cutting detection latency by **40%** and automating **85%** of manual loan application reviews at **99.2%** accuracy.
- Reduced AWS Lambda spend by **\$22K/month** by implementing intelligent concurrency and scaling policies, provisioned-concurrency tuning, and cold-start optimizations.
- Built CI/CD pipelines with GitHub Actions, Jenkins, and AWS CodeDeploy incorporating Snyk security scanning and automated deployment validation, catching **98%** of vulnerabilities pre-production.

- Instrumented production observability with Prometheus, Grafana, ELK, New Relic APM, and AWS X-Ray, and validated resilience through Gremlin chaos experiments that surfaced and remediated **3 critical failure points**, reducing MTTR by **30%**.

Software Engineer | Insight Global, India

Nov 2021 – Aug 2024

- Scaled a financial transaction management system using Angular, Java, and Spring Boot, improving platform performance to handle **50K+** daily transactions while minimizing latency for customers and financial advisors across a base of **2M+** monthly active customers.
- Optimized API response times by **40%** through Redis caching strategies, database indexing, query tuning, and connection pooling across MySQL and PostgreSQL, and orchestrated database sharding for the customer accounts system to absorb **3x** user-base growth without degrading read/write performance.
- Led a team of **4 engineers** migrating monolithic services to containerized microservices on Docker, Kubernetes, and AWS EKS, cutting deployment time by **60%** and reducing cloud infrastructure costs by **\$18K/month** while improving fault tolerance.
- Integrated Kafka-based event streaming pipelines processing **8K+** events/sec for real-time fraud alerting and transaction monitoring, reducing false positives by **25%** for security operations while enabling asynchronous communication across backend and reporting systems.
- Developed a financial anomaly detection module using Node.js and rule-based scoring algorithms, alongside a real-time cross-border regulatory compliance checker built on ML models, preventing **15%** revenue loss from fraudulent transactions and blocking **200+** potential compliance violations monthly.
- Designed a secure financial advisor-client chat feature using Angular, WebSockets, and RxJS state management, increasing advisory service engagement by **20%**, and A/B tested UI components to lift conversion on targeted campaigns by **12%**.
- Refactored payment processing gateway integrations to support **5+** new providers with idempotency, retry, and reconciliation handling, boosting transaction success rates by **18%**.
- Engineered an automated daily transaction reconciliation system using Spring Batch, processing **15K+** daily transactions at **99.9%** accuracy and eliminating manual settlement effort.
- Pioneered GraphQL adoption for mobile APIs, reducing payload size by **45%** and improving app performance metrics by **30%** through query optimization and schema design.
- Automated CI/CD pipelines using Jenkins, Groovy scripts, Git, and Docker, reducing manual deployment errors by **90%**, and improved production reliability with Resilience4j circuit breakers, retries, timeouts, and fallback patterns.
- Mentored **3 junior engineers** through code reviews, system design walkthroughs, and pairing within Agile/Scrum delivery, accelerating feature delivery by **30%** across cross-functional teams.

Software Developer | Flipkart, India

Mar 2019 – Oct 2021

- Supported **50M+** monthly active users and **20K+** requests/sec during peak sale events while reducing API latency by **30%**, by engineering Java/Spring Boot microservices for product catalog, checkout, inventory, and order workflows.
- Reduced transaction processing errors by **20%** for the payments team by developing a RESTful Java service that automated reconciliation matching, exception routing, and settlement exception handling.
- Improved fraud detection accuracy by **15%** and blocked **5,000+** suspicious transactions per month by enhancing the detection module with refined rule-based scoring and behavioral signal checks.
- Cut report generation time from **12s to 3s** for business analysts by optimizing PostgreSQL and MySQL queries through index tuning, execution-plan analysis, and query restructuring across large-scale catalog and order datasets.
- Decreased API latency by **200ms** for **10M+** daily users by integrating Redis caching across high-frequency catalog, checkout, and inventory endpoints with targeted TTL and eviction policies.
- Reduced duplicate transactions by **99%** by implementing idempotency checks and deduplication keys across payment processing and Kafka event pipelines handling **15K+** events/sec.
- Cut end-of-day reconciliation time by **40%** by collaborating on Spring Batch settlement jobs using chunk-based processing, restartable job steps, and automated retry handling.
- Reduced GC pauses by **18%** and improved service throughput by upgrading legacy Java 8 services to Java 11, migrating deprecated APIs and tuning G1 garbage collection parameters.
- Improved real-time notification delivery success rate to **99.8%** by redesigning the payment notification service with WebSockets, connection heartbeats, and retry-backed fallback delivery.
- Reduced inter-service latency by **120ms** by adopting gRPC with Protocol Buffers for high-frequency service-to-service communication, alongside Eureka service discovery, API Gateway routing, and Hystrix circuit breakers to prevent cascading failures during sale events.
- Identified and resolved **3 critical** performance bottlenecks pre-production by pioneering a Locust-based load-testing framework simulating peak sale-event traffic, and stabilized **15+** production defects in payment gateway services through root-cause analysis and regression coverage.
- Reduced onboarding time for **5+** external partner teams by documenting REST API contracts in Swagger/OpenAPI with request/response schemas, error codes, and versioning guidance.

EDUCATION

Master of Science, Business Analytics

University of Cincinnati, Cincinnati, OH | GPA: 3.85 / 4.0

Dec 2025

Bachelor of Technology, Civil Engineering

Bapatla Engineering College, India | GPA: 3.2 / 4.0

Apr 2018